

In Class Quiz

Name : _____

October 10

1. True or False

- (a) A function is differentiable if it is continuous.
- (b) One benefit of proportions is to eliminate units.
- (c) APR stands for Annual Percentage Rate.

2. Find the sum of the following series:

$$8 - 2 + \frac{1}{2} - \frac{1}{8} + \dots$$

3. What is $\ln(1.01)$ approximately?

4. The volume of timber available for harvest in a forest at time t is given by the differentiable function $f(t)$ which satisfies the following conditions:

- (a) $f(0) = 0$
- (b) $f(t)$ is convex and increasing for $0 \leq t \leq a$
- (c) $f(t)$ is concave and increasing for $a \leq t \leq b$
- (d) $f(t)$ is decreasing for $t \geq b$

Sketch $y = f(t)$